2025 MCM Junior Group Problem: Game on Integers

Jane invites Alice and Bob to play a game for fun.

First, Alice is asked to pick an integer X_1 between 1 and 100, both inclusive. Then Bob is asked to pick another integer X_2 between 1 and 100, both inclusive. X_2 must be different with X_1, and X_1 evenly divides X_2 or X_2 evenly divides X_1. Next, Alice will continue to pick an integer X_3 between 1 and 100, different from X_1 and X_2, such that X_3 evenly divides X_2 or X_2 evenly divides X_3. The game ends if the current player cannot find the number satisfying these properties.

Finally, we obtain a chain of numbers, $X_1, X_2, X_3, \ldots$, with the following properties: These numbers are between 1 and 100, with no repetition, such that if x and y are consecutive numbers in the chain, then x evenly divides y or y evenly divides x.

Jane is generous enough to pay Alice a reward equal to the sum of the numbers she has picked, that is, $X_1 + X_3 + \ldots$. Same for Bob. He will receive a reward equal to $X_2 + X_4 + \ldots$.

Here is an example of such a chain, with length 12:

$$37, 74, 2, 8, 4, 16, 48, 6, 3, 9, 27, 81.$$

Alice will receive a reward equal to $37 + 2 + 4 + 48 + 3 + 27$, while Bob receives $74 + 8 + 16 + 6 + 9 + 81$.

(1) Suppose Alice and Bob are good friends. They aim to maximize the **total reward** received. What is the chain of numbers they should choose, and what is the corresponding total reward? (10 points)

(2) (Bonus question) Suppose Alice and Bob are competitors. Alice wants to maximize her own reward and receive more money than Bob at the end. Bob wants to maximize his reward, too. Recall that Alice moves the first. If you are Alice, what is your best strategy to play this game? (2 points)

Instructions

- **Rules**: You are allowed to use any AI tools, such as ChatGPT, Gemini, and DeepSeek. For example, you can ask them to write code for you. The first question has a higher point of 10. Hence, we suggest you focus on the first one. The second question is a bonus question, which in fact is harder.

  Although this strategic game is easy to understand, it is non-trivial at all. The answer may be open-ended and rely on sophisticated tools. We encourage you to explore different ideas, show us your talent, and enjoy the pleasure of discovery.

- **Submission**: **Each team should submit a short report (no more than 4 pages) in English explaining your answers and ideas. For example, in the report, you can give the chain of numbers you find and explain how you find it. If there are any source codes used, please also attach them. In total, please send the report and code (if any) to the Email: mcm@hkust-gz.edu.cn before October 20, 23:59.**

  Remark: Each team only has to ask the team leader to send the email. Please remember to list the names of all members in your report.

- **Copyright**: Each team retains the copyright of their work. We will only use your submissions to decide the winner.

- **Why this game?** It is inspired by a puzzle from a financial firm specialized in market making. Indeed, Jane is the regulator who wants to increase the trading activities on her platform. Alice and Bob are main traders. Jane wants to know the outcomes when traders collaborate or compete with each other on her platform.

- **Academic Integrity:** This competition, like all other competitions, functions best when the participants treat one another with honesty, fairness, respect, and trust. The organizers hold all participants of this competition to high standards of scholarship and integrity.

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